

Combustion

COMBUSTION IS A REACTION THAT PRODUCES HEAT AND LIGHT IN THE FORM OF FLAMES AND EXPLOSIONS.

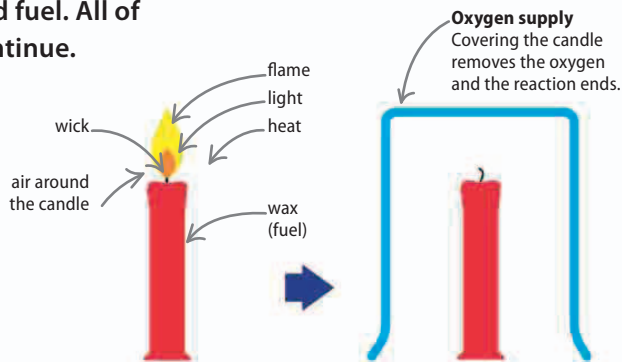
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Most combustion reactions involve oxygen, heat, and fuel. All of these components are needed for the reaction to continue.

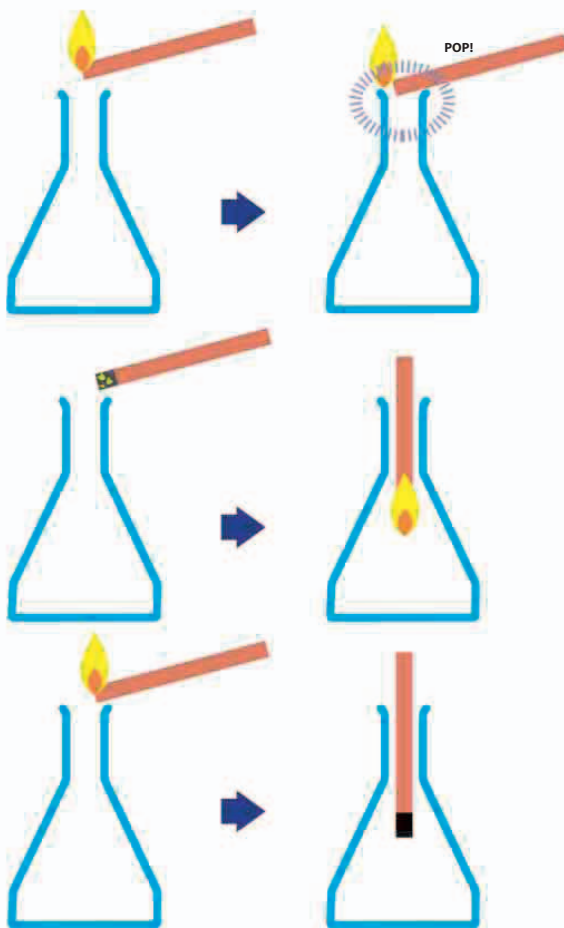
Heat and light

A flame is an area of hot glowing gases that have been released by a combustion reaction. For example, a candle wick is soaked with hot liquid wax that is a fuel and burns (undergoes a combustion reaction with oxygen in the air). The products of the reaction are carbon dioxide gas and water vapor. These glow briefly as they are released, contributing to the flame.



Gas tests

The gases commonly produced in chemistry experiments often look exactly the same. It may be dangerous to smell them—even if they have a characteristic odor. Chemists use combustion tests to identify the three most easily confused gases—hydrogen, oxygen, and carbon—in a safe way. A sample of each gas is exposed to a burning splint—a strip of dried wood used in a lab—and the gas can be identified by the characteristic way it combusts.



< Hydrogen

Hydrogen is very flammable and burns very quickly. A burning splint will pop before it even enters the flask as the hydrogen rushes out to the flame.

< Oxygen

Oxygen is the gas that fuels combustion. If a smoldering splint is exposed to a flask filled with oxygen, the wood will reignite and burst into flames.

< Carbon dioxide

Carbon dioxide is a common product of combustion reactions but it does not burn itself. A burning splint will go out when it is exposed to carbon dioxide.

Combustion—and its fire—was the first chemical reaction that humans learned to control.